ANSWER ALL QUESTIONS

INSTRUCTIONS

Select the correct ANSWER and **circle the letter corresponding to the correct answer in PEN**. For questions in part II involving calculations, you are required to show detailed work on the question paper to arrive at the answer you have selected.

Part I

1. Diesel engines are classified as reciprocating internal combustion engines because they __.
   (a) use a continuous combustion process to impart rotary motion to the pistons
   (b) use energy from fuel burned outside their cylinders
   (c) burn fuel in a closed chamber which imparts linear motion to pistons
   (d) burn fuel in a combustion chamber that moves back and forth

2. Opposed-piston diesel engines are classified as _____.
   (a) two-stroke cycle single acting
   (b) two-stroke cycle double acting
   (c) four-stroke cycle single acting
   (d) four-stroke cycle double acting

3. The small end of the connecting rod is attached to the piston with a _____.
   (a) torque bushing
   (b) wrist pin
   (c) sliding wedge
   (d) crank pin

4. The use of push rods becomes necessary in a diesel engine when _____.
   (a) the camshaft is located some distance below the valve gear
   (b) the rocker arms are pivoted near their centers
   (c) two or more valves must be opened and closed at the same time
   (d) hydraulic valve lash adjusters are used

5. In a modern two-stroke, slow-speed, internal combustion diesel engine, what is the part of the engine that houses the crankshaft?
   (a) bedplate
   (b) sump
   (c) cylinder block
   (d) frame

6. In comparison to exhaust valves, intake valves of diesel engines may be fabricated from low-alloy steels because _____.
   (a) the bevelled edges of the intake valves provide for self-centering during seating
   (b) the intake valves utilize stellite-coated valve seat inserts which reduce wear
   (c) the effective volume of air passing through intake valves is less than the effective volume of air passing through exhaust valves
   (d) the intake valves are not affected by the corrosive action of exhaust gases

7. A diesel engine crankcase ventilation system _____.
   (a) prevents spark generation
   (b) removes combustible gases
   (c) determines the level of combustible gases
   (d) provides inert gas generation in the crankcase

8. A mixture of 45% cetane and 55% alpha-methyl-naphthalene is found to have the same ignition delay as a sample of diesel oil. The sample can be described as having a/an _____.
   (a) cetane number of 55
   (b) cetane number of 45
   (c) octane number of 55
   (d) octane number of 45

9. In a naturally aspirated diesel engine, the volume of air intake is directly associated with engine _____.
   (a) compression ratio
   (b) displacement
   (c) fuel pressure
   (d) cylinder clearance volume

10. One of the factors limiting the amount of load which can be put on a modern marine diesel engine is the _____.
   (a) governor sensitivity
   (b) exhaust temperature
   (c) fuel injection pressure
   (d) speed of the camshaft

11. What may cause a diesel engine cylinder head to crack?
   (a) scale on cooling passages
   (b) a leaking oil control ring
   (c) overheated intake valves
   (d) heat transfer from exhaust valves

12. The process of admitting air into the combustion space is known as _____.
   (a) scavenging
   (b) charging
   (c) pressing
   (d) cooling

13. Engines of different cylinder dimensions, power, and speed are compared on the basis of ____.
   (a) maximum pressure
   (b) specific fuel consumption
   (c) mean effective pressure
   (d) displacement volume

14. In a four-stroke IC engine camshaft rotates at _____.
   (a) same speed as the crankshaft
   (b) twice the speed of the crankshaft
   (c) half the speed of the crankshaft
   (d) none of the above

15. Compression ratio in diesel engines is of the order of _____.
    (a) 5–7
    (b) 7–10
    (c) 10–12
    (d) 14–20

16. Main advantage of a two-stroke engine over a four-stroke engine is _____.
    (a) more uniform torque on the crankshaft
    (b) more power output for the cylinder of the same dimensions
    (c) absence of valves
    (d) all of the above

17. Engines used for ships are normally _____.
    (a) four-stroke SI engines of very high power
    (b) two-stroke CI engines of very high power
    (c) four-stroke CI engines of high speed
    (d) two-stroke SI engines of high power

18. The air standard Diesel cycle consists of an _____.
    (a) isobaric heat addition processes
    (b) isochoric heat addition process
    (c) isothermal heat addition process
    (d) isentropic heat addition process

19. The cetane number is the index used to measure a fuel oil's _____.
    (a) antiknock characteristics
    (b) ignition quality
    (c) rate of vaporization
    (d) viscosity

20. The range of volumetric efficiency of a diesel engine is _____.
    (a) 65 – 75%
    (b) 75 – 85%
    (c) 85 – 90%
    (d) 90 – 95%

Part II

21. The mechanical efficiency of a single-cylinder four-stroke engine is 80%. The frictional power is estimated to be 30 kW. Calculate the indicated power developed by the engine.
    (a) 125 kW
    (b) 150 kW
    (c) 175 kW
    (d) 120 kW

Use the information provided below to answer questions 22 and 23

A four-stroke, four-cylinder diesel engine running at 2400 rpm develops 60 kW. The brake thermal efficiency is 30% and the calorific value of fuel is 45 MJ/kg. The engine has a bore of 120 mm and a stroke of 100 mm. Take the mechanical efficiency to be 80%

22. Determine the brake mean effective pressure
    (a) 4.28 bar
    (b) 5.16 bar
    (c) 6.63 bar
    (d) 7.42 bar

23. Determine the indicated thermal efficiency
    (a) 30.4%
    (b) 37.5%
    (c) 41.3%
    (d) 45.2%

24. A four-stroke diesel engine at full load delivers 50 kW. If the friction power is 8.5 kW, find its mechanical efficiency at full and half loads respectively.
    (a) 74.6%, 37.3%
    (b) 85.8%, 42.8%
    (c) 85.8%, 74.6%
    (d) 74.6%, 42.8%

Use the information provided below to answer questions 25 to 27

In an engine working on a Diesel cycle, the inlet pressure and temperature are 1 bar and 17 °C respectively. The pressure at the end of adiabatic compression is 35 bar. The ratio of expansion i.e., after constant pressure heat addition is 5. Take $\gamma = 1.4$ and $C_p = 1.004$ kJ/kg K.

25. Determine the cut-off ratio
    (a) 1.48
    (b) 2.54
    (c) 3.26
    (d) 4.13

26. Determine the temperature at the end of combustion
    (a) 1759.3 °C
    (b) 2032.3 °C
    (c) 1952.4 °C
    (d) 2120.5 °C

27. Determine the amount of heat supplied
    (a) 1048 kJ/kg
    (b) 1189 kJ/kg
    (c) 1236 kJ/kg
    (d) 1302 kJ/kg

28. Determine the amount of heat rejected
    (a) 432 kJ/kg
    (b) 558 kJ/kg
    (c) 614 kJ/kg
    (d) 701 kJ/kg

29. Determine the efficiency of the cycle
    (a) 42%
    (b) 48%
    (c) 51%
    (d) 55%

30. A Diesel engine is working with a compression ratio of 15 and expansion ratio of 10. Calculate the air-standard efficiency of the cycle. Assume $\gamma = 1.4$.
    (a) 42%
    (b) 57%
    (c) 63%
    (d) 71%

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